

The Nexus of Leadership and Learning: Exploring Female Principals' Influence on the Teaching-Learning Process in Secondary Schools

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Abstract: The research paper will set out to debate the role played by the leadership competencies of principals in the teaching and learning process. The research will embrace a quantitative research design that will seek to answer the question concerning the relationship that exists between leadership skills of possessed by the principals and the ongoing teaching-learning process in the female secondary schools using a co-relational study design. A survey instrument comprising of 37 items, which measured the leadership skills possessed by the selected principals and the prevailing teaching learning process, was used in the collection of data. Questionnaires for this study were administered to 108 male and female educators who were serving in secondary schools in Tehsil Wari, upper Dir and among them 100 questionnaires were submitted and were deemed to be useful in the analysis. The data in hand were calculated using Statistical Package of the Social Sciences (SPSS) version-27 and the obtained results indicated that the collected data was normally distributed. The findings revealed that all the variables of the study had great correlations. The analysis of the data was done through linear regression to measure the effect that the leadership skills of the principals had on the teaching learning process and this indicated that leadership skills of the principals have statistically significant effect on the teaching learning process. The policies derived out of this research can be a good guide to both the policymakers and organizational leaders because they can provide a good outline on how to empower the leaders in charge of the principals and as well as exert more influence on the effective teaching and learning processes.

Key Words: Principal's Leadership Skills (PLS), Teaching Learning Process (TLP), Secondary Schools (SS)

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Introduction

In contemporary society, education plays a vital role in promoting stability and economic security. Its benefits extend beyond financial protection, as literacy skills such as reading and writing enable individuals to safeguard themselves against exploitation, including avoiding fraudulent documentation. Education is therefore central to human development and is recognized as a fundamental right of every citizen of Pakistan. The quality of a nation's educational system is closely linked to its social and economic progress, and its strengthening significantly influences broader national prosperity. In accordance with the Constitution of Pakistan, the state is obligated to provide free education to children between the ages of 5 and 16. Within this system, secondary education occupies a critical position, serving as a transitional phase between elementary and higher education during which students make key decisions regarding their future academic and career paths.

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The success of any organization—whether a business, school, or other institution—is a primary concern for its leadership and management. Effective leadership is essential for organizational growth and optimal performance, as it guides behavior, clarifies goals, and determines strategies for achieving desired outcomes. Leadership within an organization function as a motivating force, as leaders inspire subordinates to work toward shared objectives (Hubbart, 2024). While the ability to anticipate and plan for the future may be an inherent quality of leadership, not all leadership styles yield positive outcomes. Authoritarian leadership, for instance, is often characterized by egocentrism and excessive concern with status. Such leaders tend to disregard the contributions of organizational members, prioritize personal or financial gain, and resist feedback or alternative viewpoints, believing their decisions to be superior.

There is a variation in teacher performance among individuals and different situations. As de Jong et al. (2022) claims, verbal and non-verbal behaviors of teachers indicate their competence, professional attitudes, and pedagogical skills. The concept of performance is typically seen as an expression of the capabilities of a person in a particular situation, such as the ability to work under different degrees of pressure. In teaching contexts, teaching performance can be defined as the various daily instructional activities that teachers use to influence the cognitive involvement and the desire to learn in students. The effectiveness of teachers is usually expressed by the choice of the right instructional strategies, the quality of the classroom interactions, and the use of non-verbal communication. Moreover, the professional identity of a teacher plays a significant role in all aspects of practice, such as the choice of instruction, classroom management, and communication with students.

This study represents a valuable contribution to the Pakistani educational context, where limited research has examined and explained the relationship between leadership skills and teacher performance. Key elements influencing teacher effectiveness include performance, management style, influence on students, and job satisfaction (Aboramadan et al., 2021). Teachers tend to perform at their highest level when they experience job satisfaction, as it enables them to express creativity and innovation in their teaching practices. Such innovative thinking benefits both teachers and students and leads to improved educational outcomes (Hsieh et al., 2024). Given the scarcity of empirical studies on leadership and teacher performance in Pakistan, this research adds significant value by identifying effective leadership and teaching practices and exploring the relationship between them. The findings are expected to benefit educators across all levels, inform principals' leadership approaches, enhance institutional management, and contribute to teacher preparation programs and secondary education practices more broadly.

Objectives of the Study

The current study focuses on the relationship of Principal Leadership Skill (PLS) and Teaching Learning Process (PLS). Specifically, the objectives of the study are as below.

1. To examine the relationship of Principal Leadership Skill (PLS) and Teaching Learning Process (PLS).
2. To determine the effect of Principal Leadership Skill (PLS) on Teaching Learning Process (PLS).
3. To assess the extent to which the collected data confirm the proposed model.

Methodology

This research paper aims to look at the effect that the leadership ability of principals has on the academic performance of the students. The survey research design was utilized to describe and report the existing conditions with the help of a structured questionnaire or interview schedule. The data collection method adopted in the study, descriptive survey, enabled the research to answer the question concerning the present state of variables used in the research.

Demographic Description

There were 100 respondents who were involved in the research. In regard to gender, 36 percent of the participants were men whereas 64 percent were women. Regarding the age distribution, the majority of participants were teachers between 25 and 30 years old (40%), then there were teachers between 31 and 35 years old (31%). Moreover, a quarter of the respondents fell in the age group of 36–40 whereas only 4% fell in the age group of above 40 years.



On professional qualification, 37 percent of the respondents had a degree in Bachelor of Education (B.Ed.) and 41 percent had a Master of Education (M.Ed.). Moreover, 13 per cent had received a Certified Teacher qualification and 9 per cent had other qualification categories.

As to experience in teaching, 7 percent of teachers had an experience less than one year, and the largest proportion of teachers had 1–10 years of experience (59 percent). In addition, 19% involved experience of 11–15 years, 10% experience of the age ranging from 16–20 years and 5% experience of over 20 years of teaching experience.

With regard to the overall academic qualification, 2 out of every 100 were undergraduates, 26 out of every 100 were graduates, 69 out of every 100 respondents had master's degree and 3 out of every 100 had other academic qualification.

Table 1

Demographic Profile of the Teachers (N = 100)

Demographic Variable	Category	Frequency (f)	Percentage (%)
Gender	Male	36	36.0
	Female	64	64.0
Academic Qualification	Undergraduate	2	2.0
	Graduate	26	26.0
	Master's	69	69.0
	Other	3	3.0
Professional Qualification	B.Ed.	37	37.0
	M.Ed.	41	41.0
	Certified Teacher	13	13.0
	Other	9	9.0
Teaching Experience	Less than 1 year	7	7.0
	1–10 years	59	59.0
	11–15 years	19	19.0
	16–20 years	10	10.0
	More than 20 years	5	5.0
Age	25–30 years	40	40.0
	31–35 years	31	31.0
	36–40 years	25	25.0
	Above 40 years	4	3.9

Instrument of the Study

A close-ended and structured research instrument (questionnaire) was administered to gather the data. The instrument was in the form of two parts, the earliest part captured the demographic data of the participants, while the second part captured the primary variables under the study, namely Principals Leadership Skills (PLS) and Teaching Learning Process (TLP).

These leadership skills of the selected principals were conceptualized in three respective sub-dimensions including 1) Behavioral Leadership Skills (BLS), 2) Organizational Leadership Skills (OLS) and 3) Transformational Leadership Skills (TLS). The dimensions have been described by the use of eighteen (18) items borrowed with the prior permission of Niqab (2015). The researchers created items that measured the ongoing teaching –process and these items were later checked by subject experts on content validity. Out of their feedback, the 30 items were narrowed by cutting them down to 19 items. The completed questionnaire had a total of 37 questions and was used to collect the data.

Results

Principal's Leadership Skills and Teaching Learning Process

Table 2 provides the descriptive statistics of eight variables, which include TLP, RS, PLS, OLS, RT, TLS, BLS and RP, and were measured on 100 respondents. The statistics involves minimum, maximum, mean,



standard deviation, skewness and kurtosis. The skew and kurtosis of the results are all within the acceptable range of ± 1.96 , meaning the results are normally distributed (López-Martín et al., 2023).

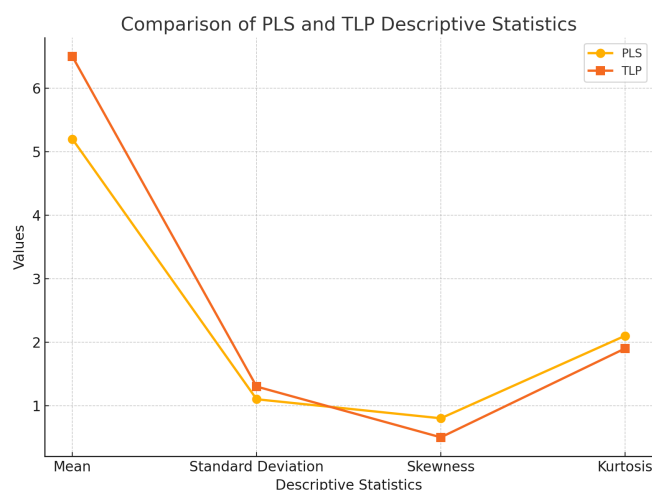
Since the observed data followed a normal distribution, it was assumed to be parametric and thus the application of the parametric statistical methods to answer the research questions. In particular, Pearson correlation analysis, as well as linear regression analysis, was used to investigate the association between variables and predictive value of principals leadership skills on the process of teaching and learning regarding the teaching and learning process. The use of these methods on normally distributed data makes these methods more robust, and the findings of the study are more credible.

Table 2

Descriptive Statistics: (N = 100)

Variable	Minimum	Maximum	Mean	Std. Deviation	Skewness	Kurtosis
TLS	1.00	5.00	3.09	0.84	-0.072	-0.238
OLS	1.00	4.60	3.25	0.93	0.161	0.455
BLS	1.00	5.00	3.22	0.87	-0.245	-0.637
RT	1.00	4.86	3.37	0.86	-0.566	-0.426
RS	1.00	4.67	3.38	0.78	-0.859	0.473
TLP	1.00	4.53	3.39	0.78	-0.056	1.663
PLS	1.00	4.83	3.16	0.76	-0.003	-0.066
RP	1.00	4.83	3.37	0.75	-0.527	0.007

Figure 1



Co-relation between PLS and its Sub Dimensions

As it is shown in Table 3, Pearson correlation coefficients between Principals Leadership Skills (PLS) and the three respective sub-dimensions of Principals Leadership Skills (i.e., 1)Transformational Leadership Skills (TLS), 2) Organizational Leadership Skills (OLS), and 3) Behavioral Leadership Skills (BLS)) has presented. The analysis of these correlations gives an idea of the interconnection between these leadership constructs and provides a clear picture of how these dimensions work together to produce efficient school leadership.

Correlation coefficients (r) can have a value between -1.0 and $+1.0$ with $+1.0$ being a perfect positive relationship, 0 being no relationship and -1.0 being a perfect negative relationship. In educational leadership studies, a positive coefficient would suggest the idea that when one dimension of leadership is enhanced, the other complementary dimensions are also enhanced. Schober et al. (2018) suggest the following interpretation of the correlation strength: $0.01-0.29$ ($p \leq .05$) means weak relationship, $0.30-0.49$ ($p \leq .05$) means moderately strong relationship, and $0.50-1.00$ means strong relationship.



The current findings reflects that the positive correlation involving Principals Leadership Skills (PLS) and Transformational Leadership Skills (TLS) is very strong ($r = 0.946$) and the sample size ($n = 100$) is significant enough ($p = 0.001$). This observation implies that transformational leadership is one of the key elements of the general effectiveness of leadership. Principals, who portray high transformational leadership (the capacity to motivate educators, communicate a clear picture, and foster a teamwork school atmosphere) will have a high probability of portraying effective overall leadership skills. The statistically significant p-value ($p < .001$) means that such an association could not take place by chance, which supports the strength of the relationship (Schober et al., 2018).

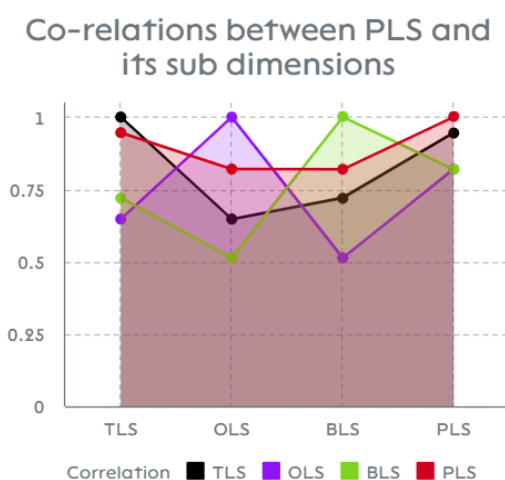
Table 3

Co-relations between PLS and Itsrepective Sub-Dimensions

Variables		TLS	OLS	BLS	PLS
TLS	Pearson -Correlation	1	0.648**	0.722**	0.944**
OLS	Pearson -Correlation	0.648**	1	0.512**	0.821**
BLS	Pearson -Correlation	0.722**	0.512**	1	0.819**
PLS	Pearson -Correlation	0.944**	0.821**	0.819**	1

**. Correlation is highly significant with the $p = 0.01$ level (two-tailed).

Figure 2



Co-relation between TLP and its Sub-dimensions

The existing associations between the ongoing Teaching and Learning Process (TLP) and its respective sub-dimensions—a) Role of Teacher (RT), b) Role of Student (RS), and c) Role of Principal (RP)—are presented in Table-4. Examining these correlations helps to understand how different roles in the educational setting interact and collectively influence the on the whole effectiveness of the teaching and learning process.

Correlation coefficients (r) recorded indicate both the direction and strength of the existing relationship between two variables, ranging from -1.0 to $+1.0$. Values closer to $+1.0$ represent a strong positive relationship, whereas values closer to -1.0 indicate a strong negative relationship. All correlations reported in Table no:4 is highly significant at the 0.001 level (two-tailed), denoted by **, suggesting a very stumpy probability that these associations occurred by chance.

As shown in Table no: 4, TLP has a strong and significant positive correlation with the significant Role of the Principal (RP) ($r = 0.873^{**}$, $n = 100$, $p \leq 0.001$). Similarly, TLP is highly correlated with the Role of Teacher (RT) ($r = 0.894^{**}$, $n = 100$, $p \leq 0.001$) and the Role of Student (RS) ($r = 0.848^{**}$, $n = 100$, $p \leq 0.001$). These findings indicate that effective ongoing teaching–learning processes depend on the active engagement and collaboration of principals, teachers, and students



Table 4

Correlations between the variable TLP & its respective Sub-Dimensions: (N = 100)

Variables	RT	RS	TLP	RP
RT	1	0.715**	0.894**	0.769**
RS	0.715**	1	0.848**	0.726**
TLP	0.894**	0.848**	1	0.873**
RP	0.769**	0.726**	0.873**	1

Note: $p \leq 0.01$ (2-tailed). RT = Role of Teacher; RS = Role of Student; RP = Role of Principal; TLP = Teaching-Learning Process.

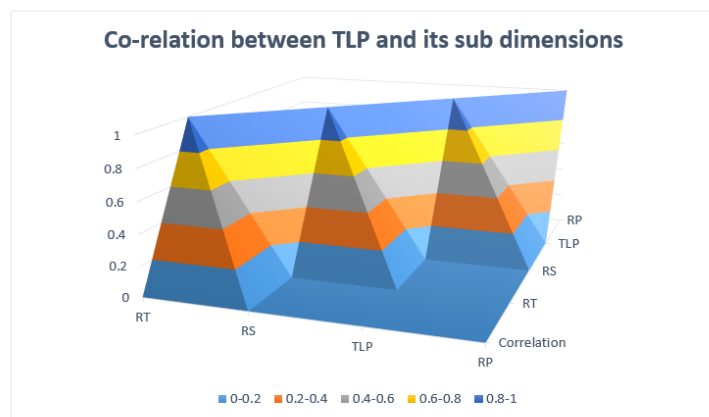
Figure 3**Co-relation between the Variables PLS and TLP**

Table no:5 shows the correlations among the Principals Leadership Skills (PLS), and the Teaching and Learning Process (TLP), and the sub-dimension of the Teaching Learning Process; a) Role of the Teacher (RT), b) Role of the student (RS) and c) Role of the Principal (RP). The knowledge of such correlations will help in gaining an idea of the relationship between leadership competencies and the contribution that teachers, students, and principals can make in achieving an effective teaching-learning environment.

The association between variables is provided in the form of correlation coefficients (r), which take values ranging between -1.0 and 1.0 . Values that are nearer to $+1.0$ represent strong positive relationships. Table-5 shows that all correlations are significant at the 0.01 level (2 tailed) with the indicators of the correlation being denoted as that of $**$ and hence they are all unlikely to have been caused by mere chance.

The outcomes indicate that there is strong and significant correlation between PLS & the TLP ($r = 0.776$, $n = 100$ and $p = 0.001$). Role of the Principal (RP) ($r = 0.740$, $n = 100$, $p = 0.001$) & Role of the Teacher (RT) ($r = 0.746$, $n = 100$, $p = 0.001$) have a strong and high correlation with PLS as well. Moreover, there is a strong positive correlation between PLS and the Role of Student (RS) ($r = 0.660$, $n = 100$, $p = 0.001$). These results indicate that leadership abilities of the principals have a strong association with the overall teaching learning process and the active involvement of teachers, students and school leadership in enhancing effective learning outcomes.

Table 5

Co-relations between Variable PLS with the Variable TLP and its Sub Dimensions (N=100)

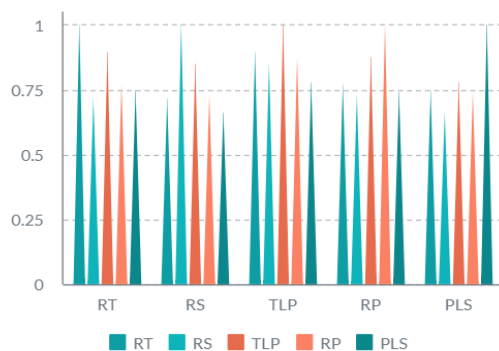
Variables	RT	RS	TLP	RP	PLS
RT	1	0.715**	0.894**	0.769**	0.746**
RS	0.715**	1	0.848**	0.726**	0.660**
TLP	0.894**	0.848**	1	0.874**	0.776**
RP	0.769**	0.726**	0.874**	1	0.740**
PLS	0.746**	0.660**	0.776**	0.740**	1

Note: $p \leq 0.01$ (2-tailed). RT = Role of Teacher; RS = Role of Student; RP = Role of Principal; TLP = Teaching-Learning Process; PLS = Principals' Leadership Skills.



Figure 4

Co-relations between PLS with TLP and its sub dimensions



Relationship between PLS and TLP

Table-6 depicts the outcome of the Analysis of Variance (ANOVA) that evaluated the correlation between the Leadership Skills (PLS) of the Principals and the Teaching and Learning Process (TLP). This examination is vital to the establishment of how many differences in the teaching learning process can be attributed to leadership competencies of principals as well as the statistical significance of this association.

ANOVA is a statistical method that is usually applied to judge the total significance of a regression model or the means of two or more groups. When applied in the context of regression, it evaluates how strongly the independent variable which in this case is PLS predicts the dependent variable, which is TLP. The most important ANOVA results are the Sum of Squares (SS), degrees of freedom (df), Mean Square (MS), the F-statistic, and the level of significance (p-value) (Howard & Knight, 2022).

Using Table 6, the p-value of the regression model is 0.000 which is less than the standard level of 0.05. This implies that the PLS and TLP have a statistically significant relationship that implies that the leading competency of the principals is a major interpreter of the effectiveness of teaching and learning process.

Table 6

ANOVA for Significant Relation (N=100)

Model	Sum of the Squares	df	Mean Square	F-value	Significance
Regression	37.454	1	37.454	148.669	0.000
Residual	24.030	98	0.245	—	—
Total	60.484	99	—	—	—

Note: Dependent Variable: TLP; Predictor: PLS

Discussion

The results of this research project can be useful in understanding how the leadership skills of female principals can affect the teaching and learning process in the selected secondary schools in Tehsil Wari, Upper Dir. The findings prove that there is a strong and significant positive association between Leadership Skills (PLS) of the Principals and the Teaching Learning Process (TLP). This correlation creates a service relationship between principals with well-developed, strategic and resourceful leadership skills to provide an educational space that encourages teachers and students to achieve competent teaching activities and improved learning results. The analyses of the data explain that transformational leadership skills (TLS) is effective in teaching-learning process that agrees with the available evidence. Including more transformational perceptions such as employee support, vision building and work group management to the principals not only improves the school environment but also hires the teachers thus resulting in better performance of the students. This finding aligns with other studies that have taken into account the issue of leadership behavior as it relates to teacher motivation and the way their teaching practice can be improved in accordance with the study (Zhang et al., 2024; Saleem et al., 2024).



Further, the paper establishes that organizational leadership skills (OLS) other than behavioral leadership skills (BLS) are also a major component in ensuring the creation of a conducive learning environment. Principals that possess good skills in organization may be capable of organizing the resources, facilitating the teaching staff, and combining the administrative tasks with the educational objectives. In the same vein, with good behavioral leadership, principals establish good relationship with the staff and hence they can easily involve the staff in a positive sharing that improves the teaching environment.

The results show that there is a close relationship of the roles that teachers (RT), students (RS) and principals (RP) play in the teaching learning process. This dependency highlights the need to have a holistic and cooperative leadership style in schools. Although the ability of teachers to fulfill the responsibility of implementing the educational policy depends on their level of learning, the level of support offered by school principals also plays a great part in dictating their effectiveness. On the same note, the academic performance of the students improves positively and in a conducive environment that is developed together by teachers and principals.

The correlated relationship between the Leadership Skill (PLS) of Principals and Teaching Learning Process (TLP) indicates that leadership cannot be considered as a periodical or structural activity in an organization but a dynamic and ongoing process that can be found in every sphere of education. Principals who perform highly are those who in addition to influencing the teaching practices have a direct effect on the learning process of the students as well as their academic outcomes. This social network underpins the thesis that leadership development initiatives ought to be part of the educational policy in place, especially in those areas like Upper Dir where the education system is in transition and where greater importance is now being placed on the role of women as leaders in the educational achievements.

In addition to this, findings of this research are consistent with international studies that prove that school leadership is multiplicative towards the performance in education. Meanwhile, the results demonstrate the exceptional situation of Pakistan and particularly female-based institutions where the practices of leadership are strongly guided by sociocultural standards. The close and positive correlation found between PLS and TLP is a good recommendation to the policymakers and education leaders who should invest in leadership development programs to teachers in the leadership positions. Such programs ought to be based on improving transformational, organizational, and behavioral leadership capabilities in order to improve the overall teaching learning environment.



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